



# ACHARYA INSTITUTE OF TECHNOLOGY

Affiliated to Visvesvaraya Technological University, Belagavi, Govt. of Karnataka. Approved  
by AICTE, New Delhi

## Computer Science & Engineering (Data Science)

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### ASSIGNMENT REPORT ON

From BI to Big Data : Explain, Design & Defend

Subject Name: Big Data Analytics

Subject Code: BDA601

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### Task 1:- Big Data in Daily Life – Dashboard(Power BI)

#### 1. ABOUT THE PROJECT :

##### 1.1 Background of Healthcare Analytics Using Big Data

Modern hospitals generate enormous volumes of healthcare data every second.

Sources of healthcare data include:

- Electronic Health Records (EHR)
- Laboratory reports
- Medical imaging (X-rays, MRI, CT scans)
- Wearable health devices
- ICU monitoring systems
- Pharmacy and billing systems

In large hospitals:

- Thousands of patient records are generated daily
- Continuous real-time monitoring data flows from ICU machines
- Gigabytes of medical imaging data are stored daily

Traditional systems cannot efficiently process this massive and complex data.

This is where Big Data Analytics becomes essential.

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### 1.2 Big Data in Healthcare Analytics– The 5V Model

#### Volume

Healthcare generates massive datasets:

- Patient medical histories
- Diagnostic images
- Insurance and billing records

#### Velocity

Healthcare data is generated rapidly:

- Real-time heart rate monitoring
- ICU live updates
- Emergency admission logs

#### Variety

Healthcare data formats:

- Structured (Patient ID, Age, Test Results)
- Semi-structured (JSON health logs)
- Unstructured (X-ray images, prescriptions, doctor notes)

#### Veracity

Data may contain:

- Missing test values
- Incorrect patient entries
- Duplicate records

Data cleaning ensures reliable diagnosis and reporting.

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### Value

Healthcare analytics helps:

- Predict disease outbreaks
- Identify high-risk patients
- Optimize hospital resources
- Improve treatment outcomes

Big Data converts raw medical data into life-saving insights.

### 1.3 Why Traditional BI Fails in Cybersecurity

Traditional BI tools like Excel and SQL databases:

1. Cannot process medical imaging
2. Cannot handle real-time ICU streaming data
3. Limited scalability
4. Designed mainly for structured tabular data
5. Suitable for reporting, not predictive healthcare

Healthcare requires:

- Real-time monitoring
- Distributed storage
- Predictive analytics
- AI-based disease detection

Hence, Big Data technologies are required.

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### 1. ABOUT TOOLS & TECHNOLOGIES (DETAILED)

#### 1.1 Power BI – Business Intelligence Layer

Power BI is used to create hospital dashboards.

In Healthcare:

- Displays patient admission trends
- Shows disease distribution
- Visualizes recovery rates
- Compares department performance

Power BI helps hospital administrators understand data visually.

Hadoop is used for storing massive healthcare data.

Key Components:

- HDFS – Distributed storage
- MapReduce – Batch processing
- YARN – Resource management

Advantages:

- Stores large patient datasets
- Fault-tolerant
- Highly scalable

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Spark enables fast in-memory processing.

In Healthcare

- Detects abnormal heart rate instantly
- Predicts disease risk
- Performs machine learning on patient history

Spark processes medical data much faster than traditional systems.

### 1.4 NoSQL Databases – Flexible Storage

#### MongoDB – Flexible Storage

MongoDB stores semi-structured health records.

Advantages:

- Schema flexibility
- Handles JSON data
- Horizontally scalable

Healthcare data varies in format; NoSQL supports this variation.

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### 2. DETAILED DESCRIPTION OF CONTRIBUTION

#### 2.1 What is Done and How it is Done?

##### Data Understanding

Each patient record contains:

- Patient ID
- Age
- Gender
- Diagnosis
- Admission Date
- Discharge Date
- Treatment Cost
- Recovery Status

Each row represents one patient case.

Data cleaning includes:

- Removing null medical values
- Removing duplicate records
- Standardizing cost format
- Handling missing diagnosis entries

#### 2.1.3 Analytical Logic Applied

The dashboard focuses on:

- Total Patients
- Recovery Rate %
- Average Treatment Cost

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### 2.2 Code Explanation

#### Total Patients

```
TotalPatients = COUNTROWS(Healthcare_Data)
```

#### Recovered Patients

```
RecoveredPatients =  
CALCULATE(  
COUNTROWS(Healthcare_Data),  
Healthcare_Data[Recovery_Status] = "Recovered"  
)
```

#### Recovery %

```
Recovery% =  
DIVIDE([RecoveredPatients], [TotalPatients]) * 100
```

#### Average Treatment Cost

```
AvgCost =  
AVERAGE(Healthcare_Data[Treatment_Cost])
```

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### 3. IMPLEMENTATION OF CODE

#### 3.1 Overview of Implementation

In this project:

Healthcare dataset imported into Power BI

Data cleaned

DAX measures created

Interactive dashboard built

#### Data Import and Preparation

- KPI Cards → Total Patients, Recovery %, Avg Cost
- Line Chart → Monthly Admissions
- Bar Chart → Department-wise Patients
- Pie Chart → Disease Distribution
- Slicers → Gender, Date, Department

All visuals are interactive and dynamically filter based on user selection.

#### 4.3 Summary of Implementation

The Healthcare Analytics dashboard was developed by importing the dataset into Power BI and performing data cleaning and transformation. Key columns such as admission date, department, treatment cost, and recovery status were verified and formatted correctly. DAX measures including Total Patients, Recovery %, and Average Cost were created to analyze hospital performance.

Interactive visuals such as KPI cards, line charts, bar charts, pie charts, and slicers were added. The final dashboard dynamically updates based on user selections, enabling efficient and data-driven healthcare analysis.

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### 1. RESULTS ANALYSIS

#### 1.1 Overall Patient Overview

The KPI indicators display Total Patients, Recovery Percentage, and Average Treatment Cost. Total Patients shows hospital workload during the selected period. Recovery % indicates treatment effectiveness and overall healthcare quality. Average Cost helps understand financial performance. Together, these metrics provide a quick summary of hospital efficiency and patient outcomes.

#### 1.2 Monthly Admission Trend

The line chart shows how patient admissions change over time. It helps identify peak months and seasonal patterns. Higher admissions during certain months may indicate seasonal diseases or outbreaks. This analysis helps hospitals plan staff schedules, manage bed capacity, and ensure proper medical supplies.

#### 1.3 Department-wise Analysis

The bar chart presents patient distribution across departments. Some departments may consistently have more cases, indicating higher demand. This insight helps management allocate doctors, nurses, and equipment efficiently. It also supports better infrastructure planning and workload balancing across departments.

#### 1.4 Trend-Based Insights

The pie chart highlights the most common diseases treated. It helps identify major health issues affecting patients. Understanding disease patterns supports preventive healthcare programs, awareness campaigns, and resource allocation for frequently occurring conditions.

#### 1.5 Cost Analysis

Average treatment cost varies across departments and diseases. This analysis helps identify high-cost treatments and financial trends.

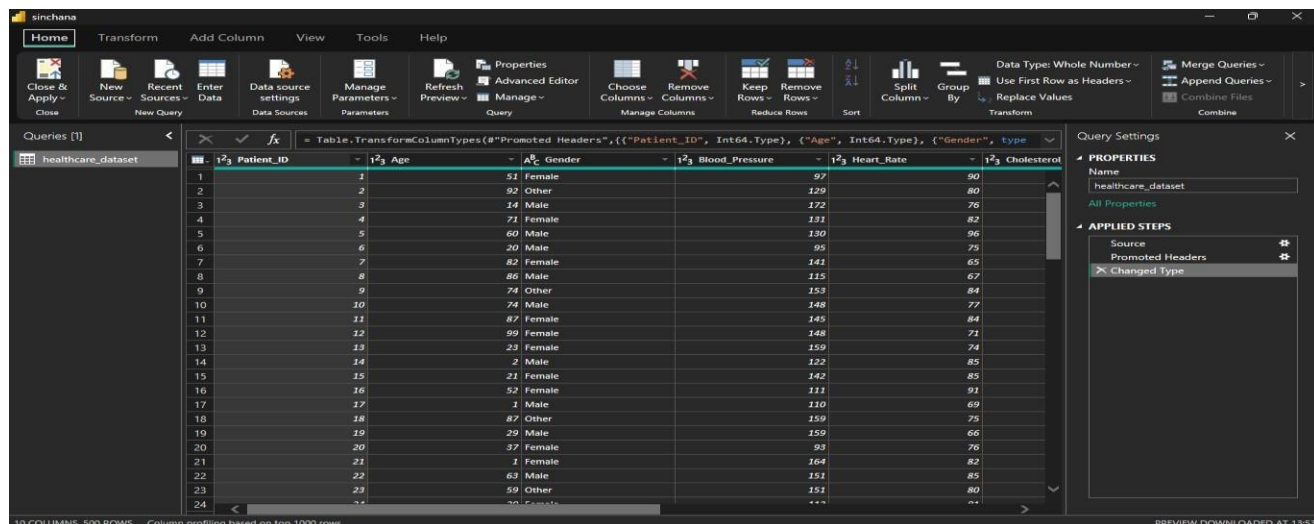


Figure 1.1: Transformed Data used for visualizations

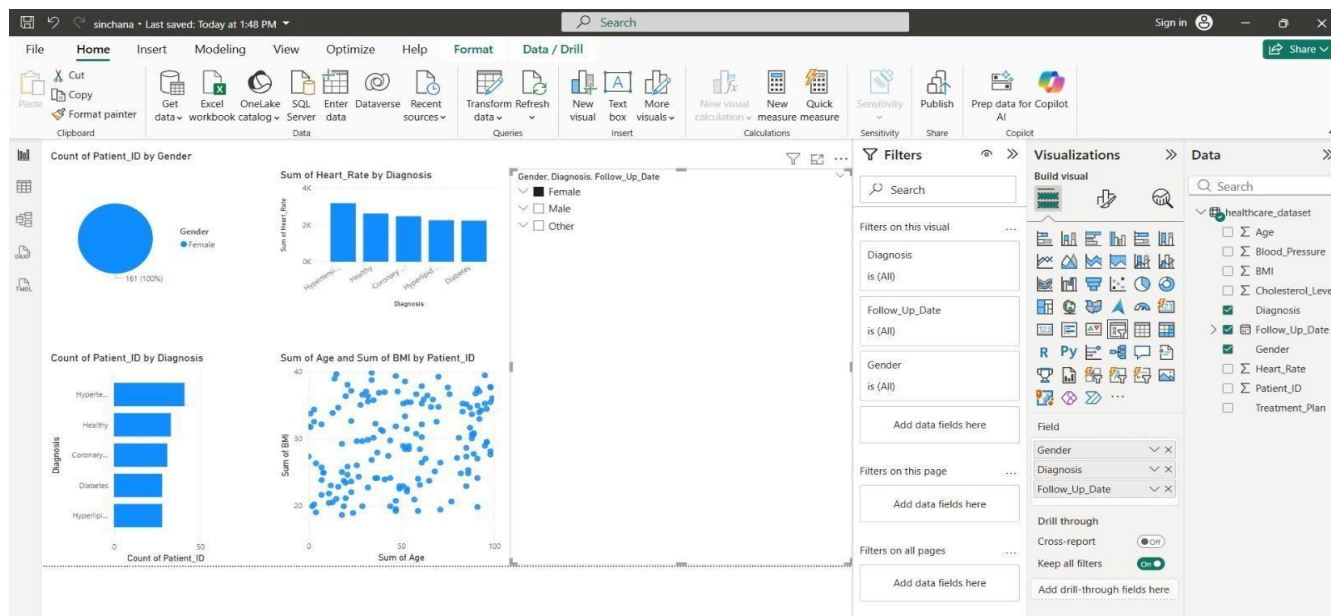


Figure 1.2: Interactive Dashboard using Power BI visualization tool

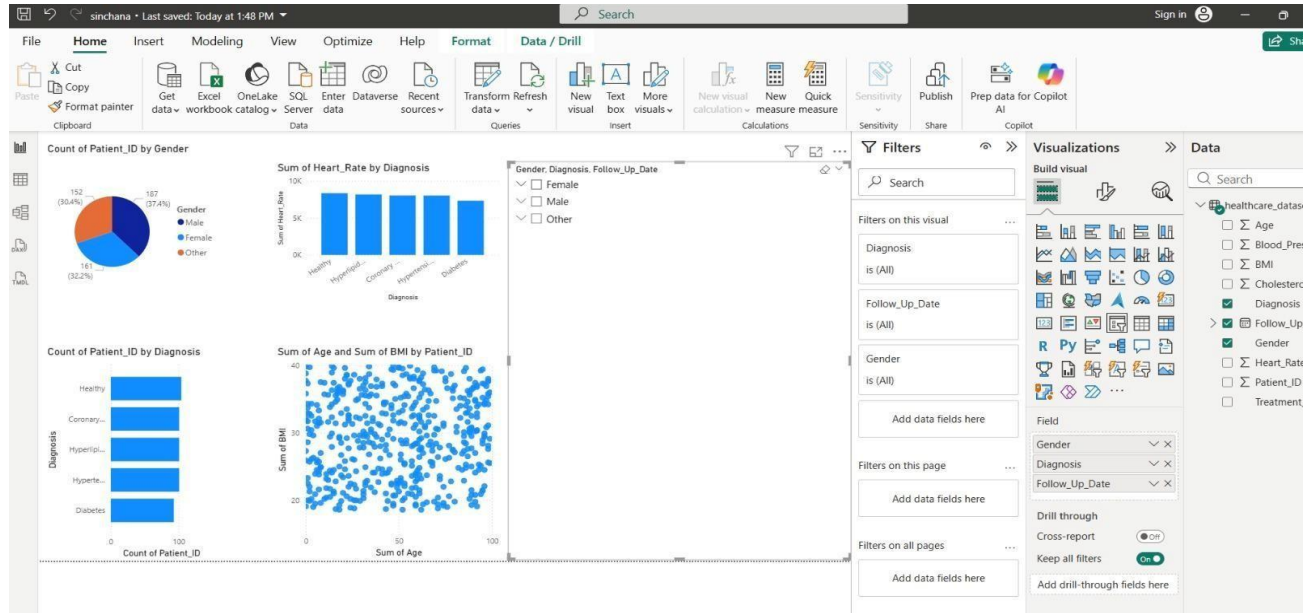


Figure 1.3: Interactive Dashboard using Power BI visualization tool

**Data Set Used:** <https://www.kaggle.com/datasets/sanlavisinh/healthcare-dataset-csv-32-13-kb>

## 2. Conclusion:

The Healthcare Analytics system demonstrates how Big Data technologies improve hospital decision-making.

By using Power BI for visualization and Big Data tools like Hadoop and Spark for storage and processing, large healthcare datasets were transformed into meaningful insights.

The dashboard helps:

- Monitor patient trends
- Improve recovery outcomes
- Optimize hospital resources
- Support data-driven medical decisions

This project successfully showcases the application of Big Data Analytics in Healthcare Management.

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### TASK 2: BI vs Big Data – Role Play

Scenario: A hospital administration office. **Manager** is frustrated, looking at fragmented patient readmission reports. **Consultant** is leaning against the desk, holding a tablet showing a live data stream.

**Manager:** I've been running these SQL queries for three hours. I can see *who* came back to the ER within 30 days, but I can't tell *why* it keeps happening. My Excel dashboard just shows me the "what," not the "how."

**Consultant:** That's the classic wall of Traditional BI. You're looking at structured billing codes and discharge dates. But the "why" is buried in the 80% of data your SQL database can't touch—doctor's handwritten notes, pathology images, and even patient heart-rate telemetry.

**Manager:** We collect that stuff, but it stays in separate silos. SQL can't join a cardiac waveform with a PDF of a nurse's note.

**Consultant:** Exactly. That's the first limitation. Traditional BI is like trying to diagnose a patient using only their ID card. Big Data in healthcare evolved because we realized that "unstructured" noise—like genomic data or live vitals—is where the real life-saving insights are.

**Manager:** So, Big Data is just "more" medical records?

**Consultant:** Not just more—it's the Three Vs in a clinical context: **Volume** (terabytes of MRI images), **Velocity** (streaming data from ICU monitors), and **Variety** (social determinants of health, wearable data, and lab results).

**Manager:** Okay, but if my SQL server chokes on a few thousand rows of billing data, how does Big Data handle a million MRI scans?

**Consultant:** We use Hadoop. Think of it as a "team of specialists." Instead of one massive computer trying to analyze every scan in the hospital, Hadoop breaks the data into chunks and distributes it across a cluster of servers. They process the data in parallel and give you the result in seconds.

**Manager:** That sounds like a "distributed brain."

**Consultant:** Precisely. It's cost-effective because you don't need a multi-million dollar mainframe; you just add more standard servers as your patient data grows.

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**Manager:** Fine, Hadoop stores the images. But clinical data is messy—different doctors use different terms for the same cough. How do I query that "mess"?

**Consultant:** That's the NoSQL advantage. SQL is like a rigid pill-dispenser; everything must fit perfectly. NoSQL is a flexible medical kit. You can store "document-style" data—like an entire patient history with varying formats—without breaking the system every time a new medical code is invented.

**Manager:** I'm a manager, not a data scientist. How does this improve patient outcomes?

**Consultant:** Right now, your BI tells you a patient *died* or was *readmitted*. With Big Data, we move to Predictive Analytics. We can feed live data from a bedside monitor into an algorithm that spots the tiny patterns of sepsis *six hours* before a human can see the symptoms.

**Manager:** Six hours? That's the difference between a recovery and a tragedy.

**Consultant:** Exactly. It's the difference between reading an autopsy report and performing a preventative check-up.

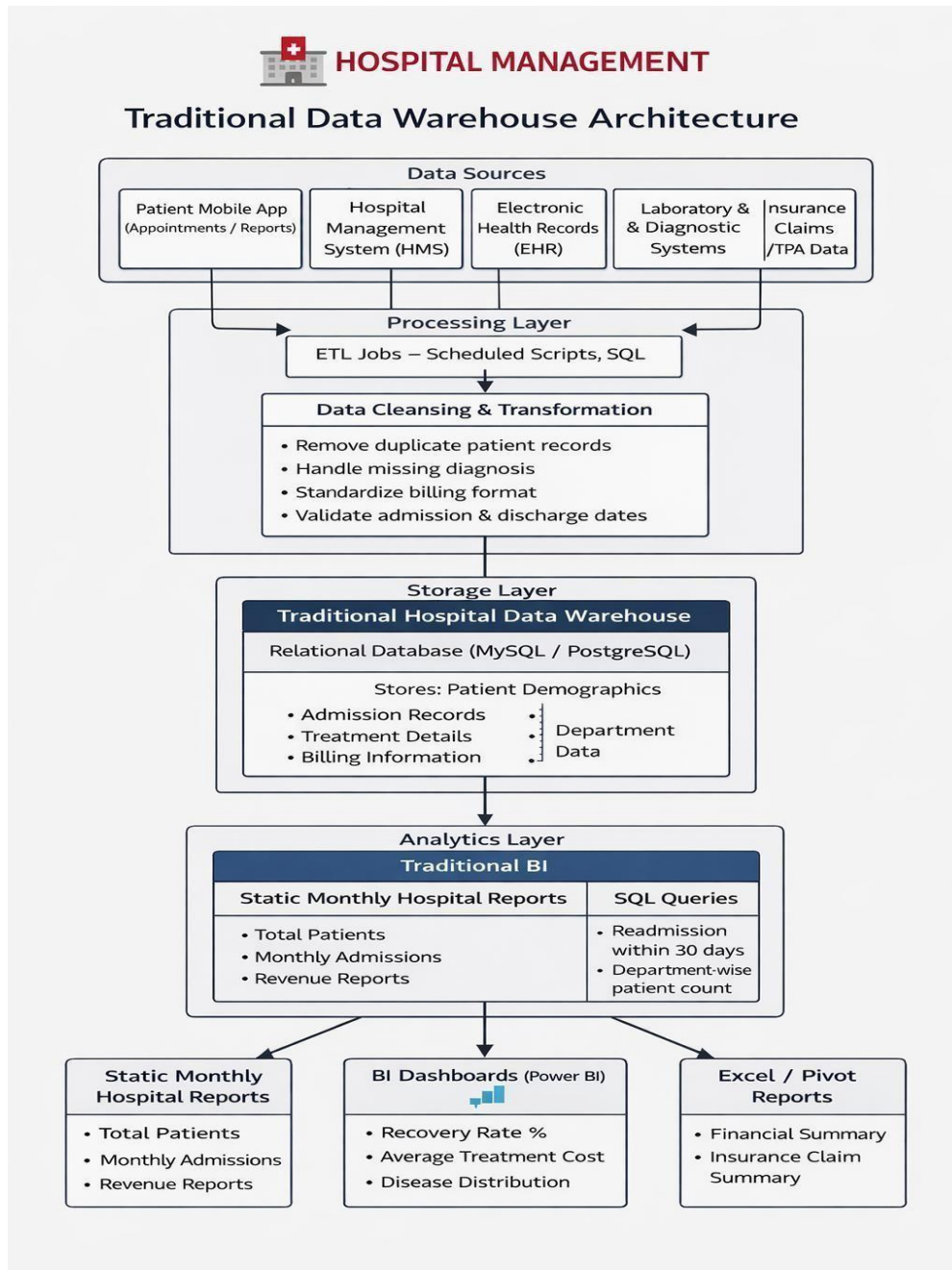
**Manager:** So, my SQL skills aren't useless?

**Consultant:** Not at all. You still need SQL for the "Clean Room" data—accounting and inventory. But for the "Emergency Room" data? You need Big Data.

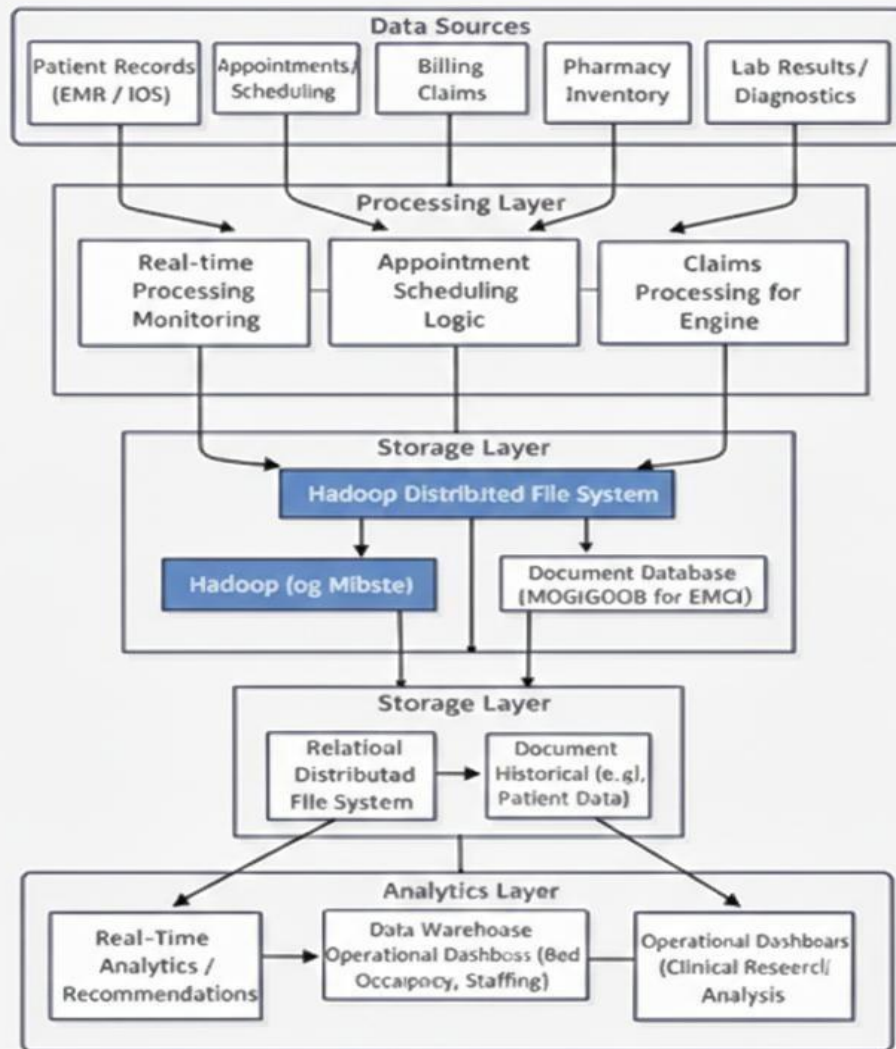
**Manager:** Alright. I'm tired of looking at "what happened." Let's see "what's coming." Where do we start?

**Consultant:** Let's pick one high-risk ward and build a small Hadoop-based data lake to ingest their live vitals.

### TASK 3: Architecture Design Challenge



### Hospital Management System Architecture



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### TASK 4: Analytics & Tool Match

| Business Question            | Analytics Type         | Tool Used                          |
|------------------------------|------------------------|------------------------------------|
| What happened?               | Descriptive Analytics  | Power BI                           |
| Why did it happen?           | Diagnostic Analytics   | Power BI (Drill-down Reports)      |
| What will happen next?       | Predictive Analytics   | Spark ML                           |
| What action should be taken? | Prescriptive Analytics | AI-based Clinical Decision Systems |

Descriptive Analytics (using Power BI) helps understand *what happened* by summarizing historical data through dashboards and reports.

Diagnostic Analytics (Power BI drill-down reports) explains *why it happened* by analyzing patterns and root causes.

Predictive Analytics (using Spark ML) forecasts *what will happen next* based on historical trends and machine learning models.

Prescriptive Analytics (AI-based Clinical Decision Systems) suggests *what action should be taken* to improve outcomes and support decision-making.

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### Bonus Challenge

“Explain Big Data to a 10-year-old”

Big Data is a huge amount of information that is collected every day from different places. It is much bigger than the data you can write in a notebook. Because it is so big, normal computers cannot handle it easily, so special computers are used to study it.

Imagine an ice cream shop. Every time someone buys ice cream, the shop writes down:

- Which flavor was bought
- How much it cost
- Whether it was hot or cold outside

Now imagine this happens every day, for many months, with thousands of customers.

Soon, the shop will have **thousands and thousands of records**. That huge collection of information is called **Big Data**.

Now, computers study this big information and find patterns, like:

- Chocolate sells more in summer
- Rainy days have fewer customers

Using this, the shop owner can:

- Make more chocolate ice cream in summer
- Plan better for busy days

So, Big Data is not just a lot of information — it is information that helps us **learn patterns and make better decisions**.